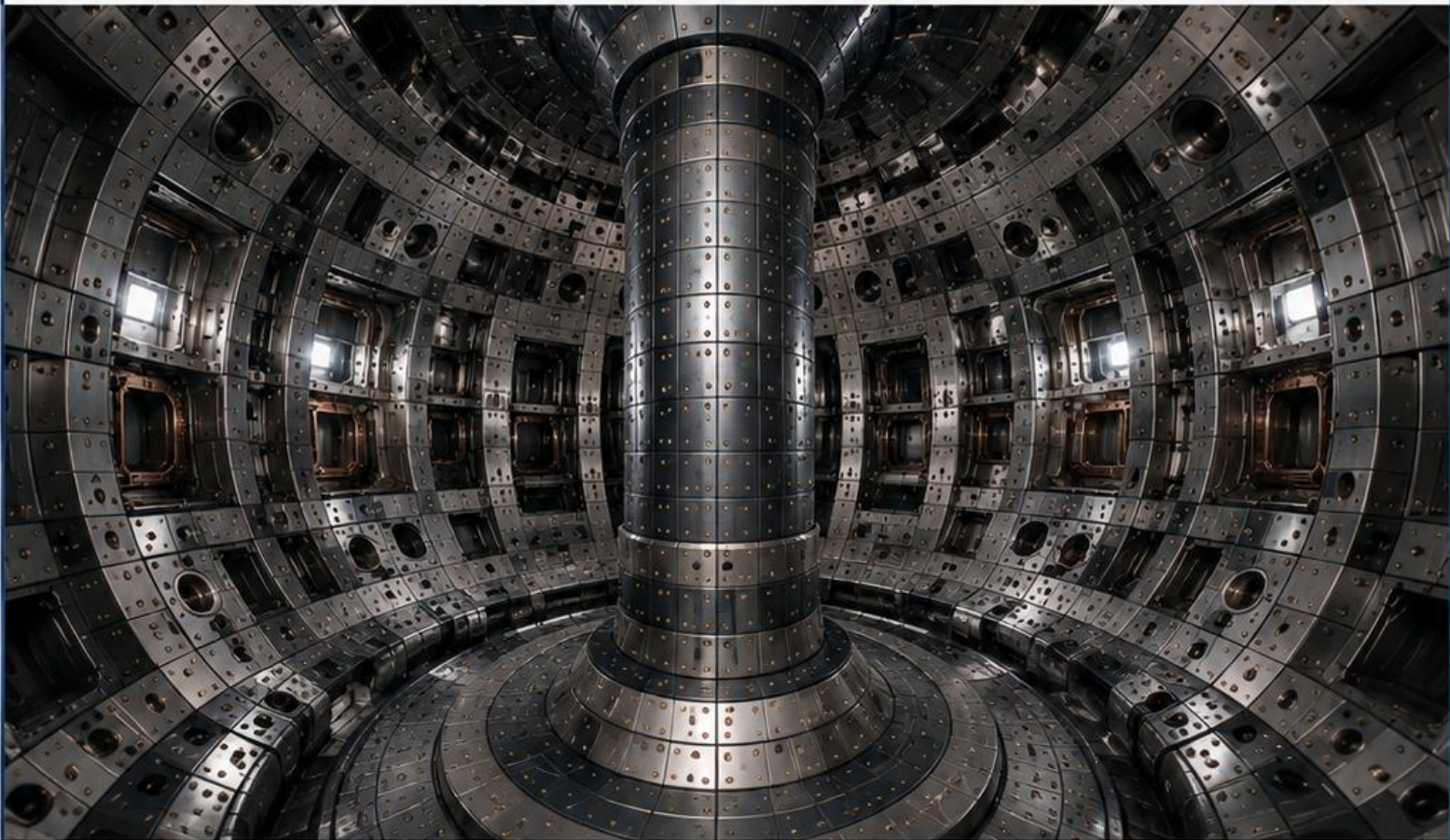


MAST-U FUSION DIAGNOSTIC DATA ANALYSIS

ANALYSIS OF MAST-U SHOT 27643

An independent analysis of plasma current, neutral beam injection power
and neutron diagnostics with uncertainty quantification.



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Contents

List of Figures	3
List of Tables	3
List Of Equations	4
Abbreviations and Acronyms	4
Introduction	5
Purpose	5
Background	5
Methodology	6
Data Acquisition and Inspection	6
Data Preparation and Time Alignment	7
Statistical and Time-Series Analysis	8
Uncertainty and Monte Carlo Analysis	9
Results	11
Data Preparation and Alignment Outcomes	11
Diagnostic Correlations and NBI Phase Comparison	13
Temporal Correlation and NBI Step Response	15
Neutron Measurement Uncertainty and Monte Carlo Analysis	17
Discussion	24
Interpretation of Diagnostic Relationships	24
Temporal Behaviour and Response	25
Robustness of Derived Metrics	25
Limitations and Assumptions	26
Conclusion	26
References	28
Appendix A — Dataset and Variable Details	29
Appendix B — Raw Data Inspection and Quality Assessment	30
Appendix C — Supplementary Statistical and Temporal Analysis	32

List of Figures

Figure 1: MAST Shot 27643 – Cleaned Analysis Window	11
Figure 2: MAST Shot 27643 – Time-Aligned Diagnostic Signals	12
Figure 3: Pearson correlation matrix for plasma current, NBI power and neutron rate within the aligned analysis window	13
Figure 4: Relationship between NBI power and neutron rate for the temporally aligned observations from MAST Shot 27643.....	14
Figure 5: Mean neutron rate during the two identified NBI operating phases. Phase 1 corresponded to a mean NBI power of 1.797 MW and Phase 2 to a mean NBI power of 3.182 MW.....	15
Figure 6: Normalised cross-correlation for NBI power–neutron rate (left) and plasma current–neutron rate (right), with peak correlations at –92.0 ms and 0 ms, respectively.	16
Figure 7: NBI step transition and corresponding neutron-rate response, showing the 3σ threshold and apparent 6.2 ms response delay	17
Figure 8 Relative neutron-rate measurement uncertainty across the 0.00–0.50 s analysis window	18
Figure 9: Neutron-rate time series with the associated measurement uncertainty envelope	18
Figure 10: Monte Carlo distribution of the measured peak neutron rate with the 95% simulation interval	20
Figure 11: Measured neutron-rate time series with the point-by-point Monte Carlo mean and 95% simulation envelope.	21
Figure 12: Monte Carlo distribution of the percentage increase in mean neutron rate from NBI Phase 1 to Phase 2.....	22
Figure 13: Distribution of maximum neutron rates obtained from the full time-series Monte Carlo simulations	23
Figure 14: Summary of full time-series Monte Carlo results.	23
Figure 15: Raw plasma-current, NBI-power and neutron-rate signals for MAST Shot 27643 prior to restriction to the analysis window.....	30
Figure 16 Raw plasma-current signal for Shot 27643.:.....	30
Figure 17: Raw NBI-power signal for Shot 27643.	31
Figure 18: Raw neutron-rate signal for Shot 27643.	31
Figure 19: Scatter plot of aligned plasma current against neutron rate.....	32
Figure 20: Normalised cross-correlation of (a) NBI power and neutron rate and (b) plasma current and neutron rate.	32

List of Tables

Table 1: Diagnostic datasets selected for the analysis.....	7
Table 2: Initial profile of the diagnostic signals.	7
Table 3: Diagnostic observations retained following application of the 0.00–0.50 s analysis window.	12
Table 4: Comparison of the two identified NBI operating phases	15
Table 5: NBI step-response comparison.....	16
Table 6: Absolute and relative uncertainty values were derived from the ANU error signal associated with the neutron-rate diagnostic.	19
Table 7: NetCDF diagnostic data sources used for Shot 27643.....	29
Table 8: Diagnostic signal metadata.....	29

List Of Equations

Equation 1: relative change in mean neutron production	8
Equation 2: Three-standard-deviation neutron-response threshold	9
Equation 3: Relative neutron measurement uncertainty	9
Equation 4: Monte Carlo generation of a simulated neutron-rate measurement	10
Equation 5: Point-by-point Monte Carlo simulation of the neutron-rate time series	10

Abbreviations and Acronyms

Abbreviation / Symbol	Definition
AMC	MAST diagnostic signal designation used for plasma current data
ANB	MAST diagnostic signal designation used for Neutral Beam Injection power data
ANU	MAST diagnostic signal designation used for neutron data
ANU_ERRORS	ANU signal containing the supplied neutron error data
ANU_NEUTRONS	ANU signal containing neutron-rate data
CI	Confidence Interval
D–D	Deuterium–Deuterium
EP	Energetic Particle
FIDA	Fast-Ion D-Alpha
Inf	Infinity / Infinite numerical value
(I_p)	Plasma current
kA	Kiloampere
MAST	Mega Ampere Spherical Tokamak
MAST-U	Mega Ampere Spherical Tokamak Upgrade
MC	Monte Carlo
MHD	Magnetohydrodynamic
ms	Millisecond
MW	Megawatt
NaN	Not a Number
NBI	Neutral Beam Injection
NetCDF	Network Common Data Form
n/s	Neutrons per second
PDF	Probability Density Function
(r)	Pearson correlation coefficient
RMP	Resonant Magnetic Perturbation
s	Second
SD	Standard Deviation
TAE	Toroidal Alfvén Eigenmode
UKAEA	United Kingdom Atomic Energy Authority
μs	Microsecond
σ	Standard deviation (sigma)
1σ	One standard deviation from the mean
3σ	Three standard deviations from the mean
%	Percentage
95% CI	95% Confidence Interval

Introduction

Purpose

The purpose of this project was to independently develop and apply a structured scientific analysis of experimental fusion diagnostic data. The dataset in which is publicly available data from MAST-U Shot 27643, provided through the United Kingdom Atomic Energy Authority (UKAEA) Open Data archive, were used as the basis of the analysis (United Kingdom Atomic Energy Authority, 2026)

The project was undertaken to demonstrate the practical application of computational data analysis techniques to a large, multivariable scientific dataset. The analytical workflow is covered in the Methodology section of this report

Emphasis was placed on the relationship between plasma current, neutral beam injection (NBI) power and measured neutron production. The analysis also investigated the temporal response of neutron production to changes in NBI power and evaluated how the reported neutron measurement uncertainty influenced derived quantities and conclusions.

The work presented in this report represents an independent analysis performed by the author. UKAEA provided the source experimental data but did not commission, perform, validate or endorse the analysis, modelling choices or interpretations presented here.

Background

The Mega Amp Spherical Tokamak (MAST) was developed at Culham as a low aspect-ratio, or spherical, tokamak for investigating the behaviour and performance of magnetically confined plasmas. The spherical tokamak configuration uses a more compact geometry than conventional tokamaks, with MAST designed at an aspect ratio of approximately 1:4. The device succeeded the earlier Small Tight Aspect Ratio Tokamak (START) and was developed both to establish the physics basis of spherical tokamaks and to contribute to wider understanding of tokamak plasma physics (Cox, 1999)

MAST supported a broad experimental research programme addressing areas relevant to both future spherical tokamak concepts and conventional fusion devices. Research performed using MAST contributed to areas including plasma transport, confinement, stability, pedestal behaviour and the physics basis supporting ITER and potential spherical-tokamak-based future facilities (Lloyd, 2011) A major upgrade programme was initiated in 2010, ultimately leading to the development of MAST Upgrade (MAST-U) (Milnes, 2015)

Experimental operation of a tokamak is characterised using numerous diagnostic and control systems that measure different aspects of the plasma and supporting systems. Three quantities were selected as the principal focus of this project: plasma current, neutral beam injection (NBI) power and neutron production. Plasma current is a fundamental parameter in tokamak operation and forms part of the magnetic control of the plasma. On MAST, NBI provided the principal auxiliary plasma heating system, using energetic deuterium neutral beams to provide heating and current drive (Nightingale, 2001)

The experimental data used in this project were obtained from the UKAEA MAST Open Data archive, which provides public access to research data from individual MAST experiments. Shot 27643 was performed on 29 November 2011 as part of an experimental programme investigating ELM control through vertical plasma kicks. The archive contains numerous diagnostic datasets for the shot, including plasma current and coil current data (AMC), neutral beam injection data (ANB), and two neutron-measurement datasets (ANT and ANU) (United Kingdom Atomic Energy Authority, 2026)

Methodology

A structured data analysis workflow was developed in MATLAB to process and analyse the selected diagnostic data from MAST Shot 27643. The workflow consisted of four stages: data acquisition and inspection, data preparation and temporal alignment, statistical and time series analysis and uncertainty assessment using Monte Carlo simulation. Each stage was implemented as a separate process to maintain traceability between the raw source data, processed datasets and derived analytical outputs.

Data Acquisition and Inspection

Diagnostic data for MAST Shot 27643 were obtained from the UK Atomic Energy Authority (UKAEA) MAST Open Data archive. The shot was performed on 29 November 2011, and the archive provides a range of analysed diagnostic datasets in NetCDF4/HDF5 format (United Kingdom Atomic Energy Authority, 2026)

Four datasets were initially selected for inspection: amc27643.nc, containing plasma current and coil current measurements, anb27643.nc, containing neutral beam injection (NBI) data and the anu27643.nc and ant27643.nc neutron measurement datasets. These classifications correspond to those provided in the UKAEA archive for Shot 27643 (United Kingdom Atomic Energy Authority, 2026)

Following inspection, three principal time dependent variables were selected: plasma current from the AMC dataset, total NBI power from the ANB dataset and neutron rate from the ANU dataset. The associated ANU neutron error signal was retained for uncertainty analysis. ANT was inspected during the initial dataset assessment but was not used as a variable in the final comparative analysis.

An initial data profile was then generated for each signal. The number of observations, missing (NaN) and infinite (Inf) values, zero and negative values, minimum, maximum, mean, median and standard deviation were calculated. The time vectors were also examined to determine the median sampling interval of each diagnostic. This profiling stage was performed before data modification so that the characteristics of the original measurements could be documented and potential quality issues identified.

Dataset	Diagnostic	Principal variable	Unit	Role in analysis
AMC	Plasma current / coil currents	Plasma current	kA	Principal variable
ANB	Neutral beam injection	Total NBI power	MW	Principal variable
ANU	Neutron measurement	Neutron rate	n/s	Principal variable
ANU	Neutron measurement	Neutron error	n/s	Uncertainty input
ANT	Neutron measurement	—	—	Initial inspection

Table 1: Diagnostic datasets selected for the analysis

Parameter	Plasma Current	NBI Power	Neutron Rate
Observations	30,000	50,001	46,704
NaN	0	0	0
Inf	0	0	0
Zero	0	0	183
Negative	16,818	15	0
Minimum	−26.48 kA	−0.0378 MW	0 n/s
Maximum	849.9 kA	3.492 MW	1.067×10^{14} n/s
Median timestep	200 μ s	50 μ s	10 μ s

Table 2: Initial profile of the diagnostic signals.

Data Preparation and Time Alignment

Following the initial data assessment, the raw diagnostic signals were visualised against their respective time vectors to identify the period associated with the plasma discharge. This assessment showed that the experimental behaviour relevant to the analysis was concentrated within the early part of the recorded signals. A common analysis window of 0.00–0.50s was therefore selected for subsequent processing.

Data outside this interval were excluded from the principal analysis rather than treating all baseline observations as erroneous measurements. This distinction was important for plasma current, where negative measurements were present within the wider raw dataset but were predominantly associated with periods outside the active region.

Potential anomalous plasma current measurements within the selected window were additionally assessed using a median absolute deviation (MAD) approach. Observations exceeding the defined threshold were flagged for investigation rather than removed. Visual examination of the flagged region indicated that the elevated measurements followed the underlying signal behaviour and could represent genuine experimental variation. The observations were therefore retained.

Statistical and Time-Series Analysis

Following data preparation and temporal alignment, exploratory statistical and time series analyses were performed to characterise the selected diagnostic signals and investigate relationships between plasma current, neutral beam injection (NBI) power and neutron production.

Descriptive statistics were calculated for each diagnostic signal, including the mean, median, standard deviation, minimum and maximum. These statistics were used alongside time series visualisations to characterise the magnitude and variability of the measurements within the selected experimental window and to support comparison between the different diagnostic signals.

Linear relationships between plasma current, NBI power and neutron rate were evaluated using the Pearson correlation coefficient. Pairwise correlation coefficients were calculated using the temporally aligned observations and presented as a correlation matrix. Scatter plots were generated to examine the distribution of observations underlying the calculated coefficients. Correlation was used as a measure of statistical association between the diagnostic signals and was not interpreted independently as evidence of a causal relationship.

Inspection of the NBI power time series identified two distinct operating regimes within the selected experimental window. The aligned dataset was therefore divided into Phase 1 and Phase 2 based on the transition in NBI power. The number of observations, mean NBI power and mean neutron rate were calculated separately for each phase. The relative change in mean neutron production between the two operating regimes was then calculated as:

Where $\overline{N}_1$ and $\overline{N}_2$ represent the mean neutron rates during Phase 1 and Phase 2, respectively.

$$\Delta N(\%) = \frac{\overline{N}_2 - \overline{N}_1}{\overline{N}_1} \times 100$$

Equation 1: relative change in mean neutron production

Temporal relationships between the diagnostic signals were investigated using normalised cross correlation analysis. NBI power and plasma current were independently cross correlated with neutron rate across a range of positive and negative time offsets. For each comparison, the lag corresponding to the maximum normalised cross correlation coefficient was identified and converted from samples to milliseconds using the common sampling interval of 200 μ s.

The cross-correlation analysis was used to assess whether the strongest statistical correspondence between each diagnostic signal and neutron production occurred at zero lag or following a displacement. However, because cross correlation across an entire experimental window can be influenced by broad trends and changes in operating regime, the calculated lag was treated as an exploratory measure of association rather than as a direct measurement of response time.

To provide a more localised assessment, an NBI step analysis was performed around the transition between the two identified NBI operating phases. The transition time was determined from the change in NBI power, and pre and post transition windows were used to characterise the neutron behaviour. Mean

NBI power and mean neutron rate were calculated for each period, together with the standard deviation of the pre transition neutron signal.

An neutron response criterion was defined using the pre transition neutron baseline. The response threshold was calculated as:

$$N_{threshold} = \overline{N_{pre}} + 3\sigma_{pre}$$

Equation 2: Three-standard-deviation neutron-response threshold

Where $\overline{N_{pre}}$ represents the mean neutron rate before the NBI transition and σ_{pre} represents the corresponding standard deviation.

The first post transition neutron rate observation exceeding this threshold was identified, and the elapsed time between the NBI transition and threshold crossing was calculated as an apparent response delay. The maximum neutron rate following the transition was also identified, together with the elapsed time between the NBI step and the neutron rate maximum.

The threshold crossing delay was treated as an analytical response metric rather than as a direct measurement of an intrinsic plasma time constant. This distinction was maintained because the observed neutron response may be influenced by the evolution of the plasma, diagnostic characteristics, signal variability and the selected threshold criterion.

Uncertainty and Monte Carlo Analysis

Measurement uncertainty associated with the neutron rate diagnostic was evaluated using the error signal supplied within the ANU dataset. The error variable, identified in the source metadata as ANU_ERRORS, contained the same number of observations as the corresponding neutron rate signal. The absolute error associated with each neutron rate measurement was examined across the selected window. Relative uncertainty was calculated by expressing the supplied error value as a percentage of the corresponding neutron-rate measurement, as shown in Equation 3: Relative neutron measurement uncertainty

$$U_{rel}(\%) = \frac{U_{abs}}{N} \times 100$$

Equation 3: Relative neutron measurement uncertainty

Where U_{rel} is the relative uncertainty (%), U_{abs} is the supplied ANU error value (n/s), and N is the corresponding measured neutron rate (n/s).

Descriptive statistics were calculated for both the absolute and relative uncertainty. The uncertainty behaviour was also assessed independently across the two NBI operating phases that were identified.

Prior to probabilistic modelling, the metadata associated with the ANU error variable were inspected to establish the meaning and units of the supplied values. The variable was identified as ANU_ERRORS, with the label errors and units recorded as error, neutr. The corresponding neutron variable was identified as ANU_NEUTRONS, with units of neutrons/sec. However, the available metadata did not specify whether the error values represented a standard deviation, standard error, confidence interval or another uncertainty convention.

Consequently, the statistical interpretation of ANU_ERRORS could not be established directly from the available metadata. For the purposes of Monte Carlo modelling, the supplied error value at each time point was therefore assumed to represent one standard deviation (1σ) of a normally distributed measurement uncertainty centred on the corresponding measured neutron rate. Monte Carlo results were interpreted subject to this assumption rather than as a definitive characterisation of the underlying diagnostic uncertainty.

Monte Carlo simulation

An initial Monte Carlo analysis was performed on the measured peak neutron rate to evaluate the propagation of the assumed measurement uncertainty at a single point. For a measured neutron rate and assumed standard uncertainty, individual simulated measurements were generated according to:

$$N_i^* = N + UZ_i$$

Equation 4: Monte Carlo generation of a simulated neutron-rate measurement

Where N_i^* is the simulated neutron rate for iteration i , N is the measured neutron rate, U is the corresponding ANU error value treated as 1σ , and Z_i is a random value drawn from a standard normal distribution.

A total of 100,000 simulations were generated for the single point analysis. The resulting distribution was characterised using its mean, median and standard deviation, together with the 2.5th and 97.5th percentiles to define a 95% simulation interval. Exceedance probabilities were additionally calculated for selected neutron rate thresholds.

Full time-series Monte Carlo analysis

The Monte Carlo approach was extended to the neutron rate time series. 10,000 simulated time series were generated by perturbing each neutron rate measurement according to its corresponding ANU error value under the same normally distributed 1σ assumption.

For time point and Monte Carlo iteration, the simulation can be expressed as:

$$N_{i,t}^* = N_t + U_t Z_{t,i}$$

Equation 5: Point-by-point Monte Carlo simulation of the neutron-rate time series

Where N_t represents the measured neutron rate at time t , U_t represents the corresponding ANU error value, and $Z_{t,i}$ is independently sampled from a standard normal distribution.

For each simulated time series, the maximum neutron rate and its corresponding occurrence time were determined. Mean neutron rates were also calculated independently for the two NBI operating phases, and the percentage increase from Phase 1 to Phase 2 was calculated using the same relationship defined in Equation 1: relative change in mean neutron production. The resulting distributions were summarised using their means, medians and 95% simulation intervals.

A point by point 95% Monte Carlo uncertainty envelope was calculated across the complete time series using the 2.5th and 97.5th percentiles of the simulated neutron rate distribution at each time point. This enabled the uncertainty associated with the measured time dependent neutron signal to be visualised.

Results

Data Preparation and Alignment Outcomes

Application of the 0.00–0.50s analysis window isolated the experimental period from the extended baseline regions present within the raw diagnostic records. Following window selection, 2,500 of 30,000 plasma-current observations, 10,000 of 50,001 NBI-power observations and 45,757 of 46,704 neutron-rate observations were retained.

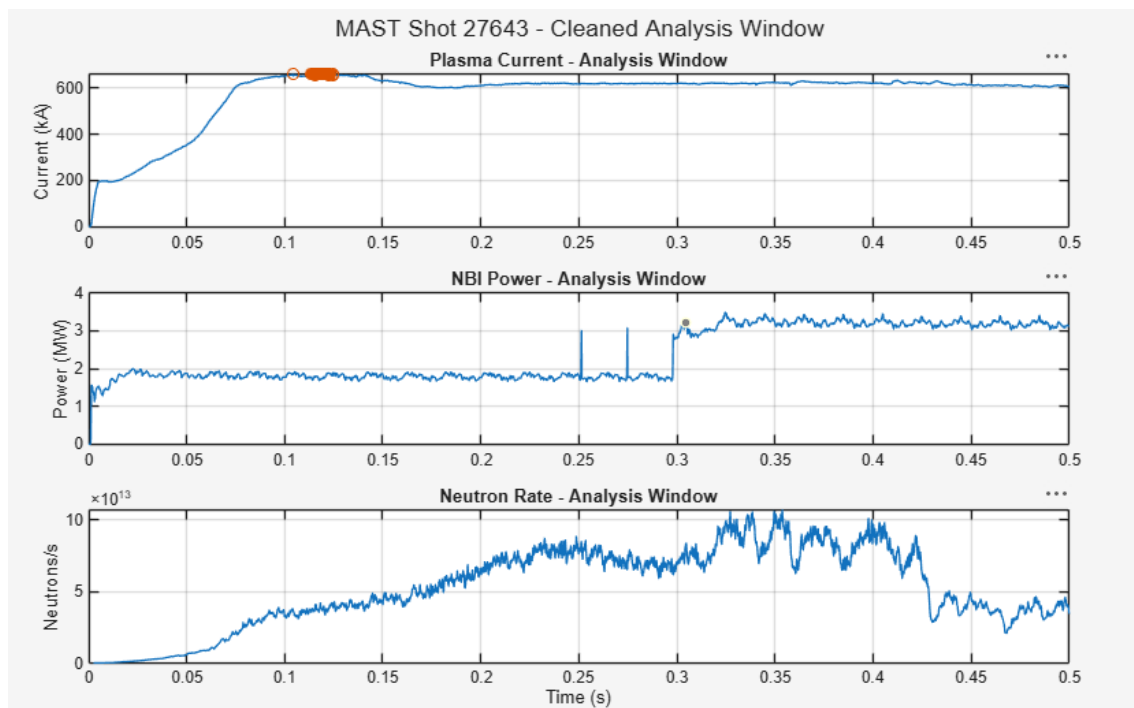


Figure 1: MAST Shot 27643 – Cleaned Analysis Window

Diagnostic	Raw observations	Retained observations	Retained (%)
Plasma current	30,000	2,500	8.3
NBI power	50,001	10,000	20.0
Neutron rate	46,704	45,757	98.0

Table 3: Diagnostic observations retained following application of the 0.00–0.50 s analysis window.

Assessment of the retained data identified a plasma current range of -2.708 to 660.044 kA and an NBI power range of -0.00824 to 3.49161 MW. Four negative plasma current observations and one negative NBI power observation remained within the selected window. The neutron rate ranged from to n/s, with no zero valued observations remaining.

Median absolute deviation (MAD) screening of the plasma current signal identified 42 observations exceeding the calculated threshold of 657.702 kA. Review of signal region indicated that these observations occurred within the elevated current behaviour of the discharge rather than as isolated deviations. They were therefore retained for analysis.

The selected diagnostic signals were aligned to a common $200\text{ }\mu\text{s}$ time base. The initial time vector contained 2,501 observations. Following interpolation, 14 rows containing incomplete values were removed, producing a final aligned dataset containing 2,487 complete observations. The resulting time series are presented in Figure 2: MAST Shot 27643 – Time-Aligned Diagnostic Signals

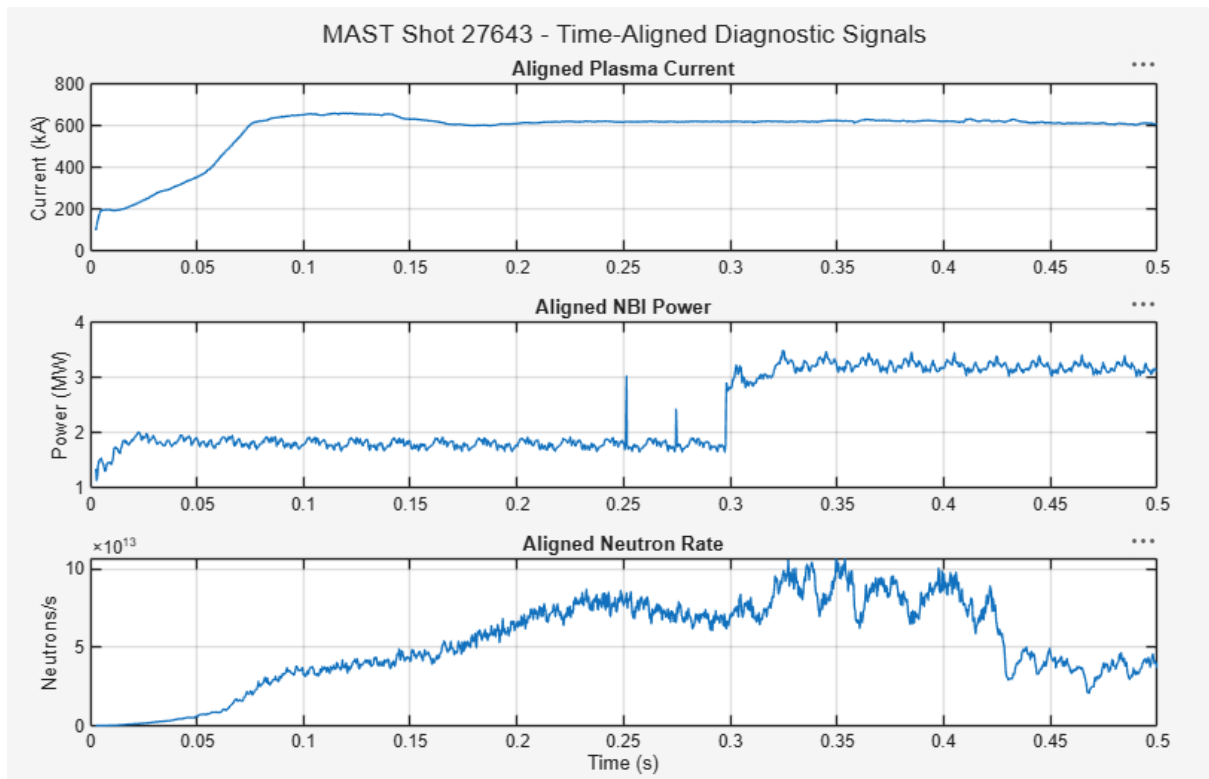


Figure 2: MAST Shot 27643 – Time-Aligned Diagnostic Signals

Temporal alignment preserved the signal behaviour observed within the selected analysis window while establishing a common set of measurement times across the three diagnostics. The resulting dataset of 2,487 complete aligned observations formed the basis of the correlation, NBI phase and temporal response analyses.

Diagnostic Correlations and NBI Phase Comparison

Analysis of the temporally aligned signals identified positive associations between plasma current, NBI power and neutron rate. The strongest pairwise relationship was observed between plasma current and neutron rate, with a Pearson correlation coefficient of $r = 0.6555$. NBI power and neutron rate exhibited a weaker positive correlation of $r = 0.4196$, while the relationship between plasma current and NBI power was weaker again, with $r = 0.3289$.

The correlation coefficients for the three diagnostic variables are summarised in Figure 3: Pearson correlation matrix for plasma current, NBI power and neutron rate within the aligned analysis window.

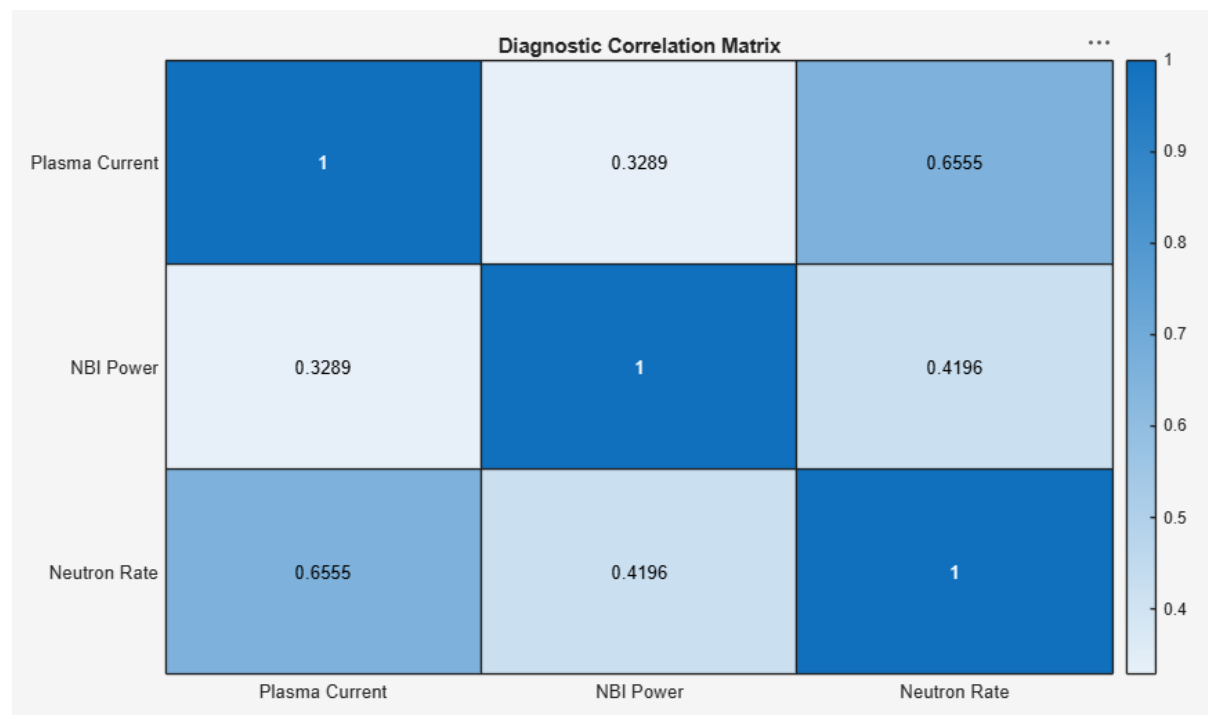


Figure 3: Pearson correlation matrix for plasma current, NBI power and neutron rate within the aligned analysis window

The correlation matrix demonstrates that all three pairwise relationships were positive, although their magnitudes differed. Plasma current showed the strongest linear association with neutron rate, whereas the lower NBI neutron correlation indicated greater variability in neutron rate at equivalent NBI power levels.

NBI Power and Neutron Rate

The relationship between NBI power and neutron production was examined further using the individual aligned observations. As shown in Figure 4: Relationship between NBI power and neutron rate for the temporally aligned observations from MAST Shot 27643, the measurements formed distinct regions associated with different levels of NBI power rather than a single uniform linear distribution.

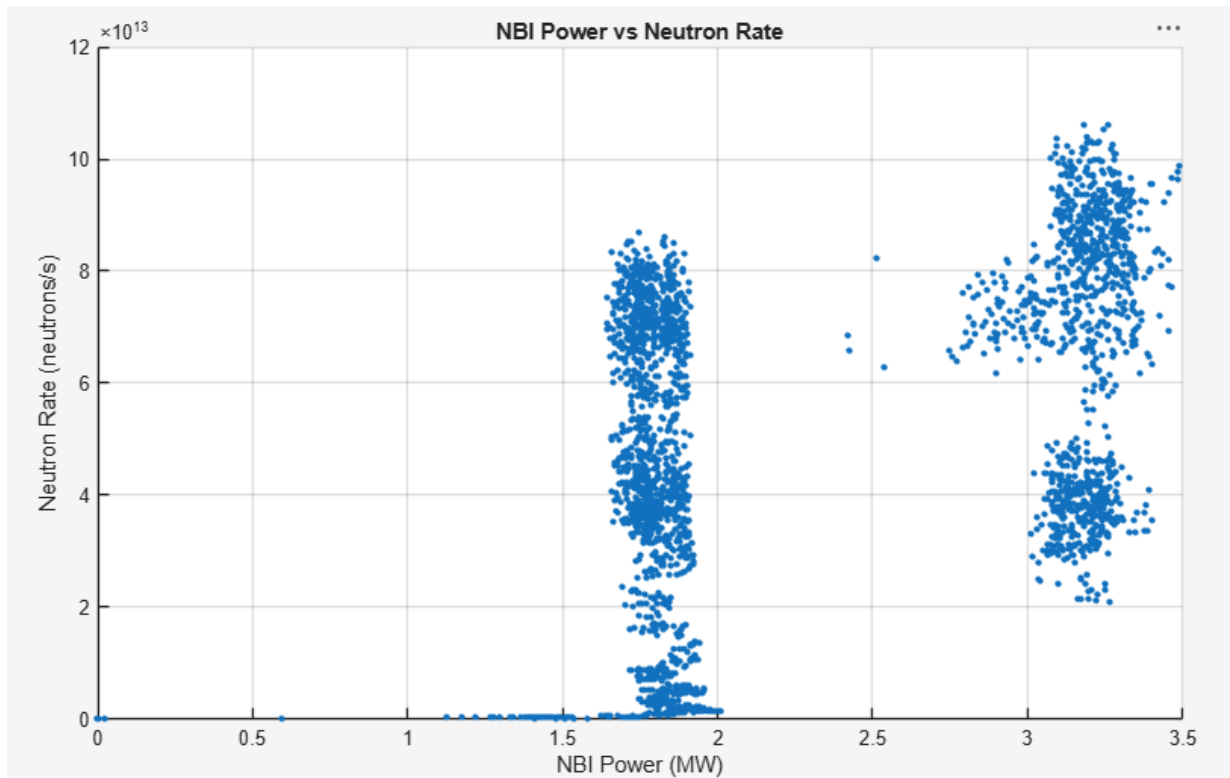


Figure 4: Relationship between NBI power and neutron rate for the temporally aligned observations from MAST Shot 27643

The distribution of observations was consistent with the presence of distinct NBI operating phases within the selected experimental window. Although higher NBI power was generally associated with higher neutron production, substantial variation in neutron rate remained within each power region. This behaviour is consistent with the moderate overall Pearson correlation coefficient and comparison of the two identified NBI phases.

NBI Phase Comparison

Separation of the dataset into the two identified NBI operating phases produced 1,441 observations in Phase 1 and 1,006 observations in Phase 2. Mean NBI power increased from 1.797 MW in Phase 1 to 3.182 MW in Phase 2, corresponding to a substantial increase in injected beam power.

Mean neutron rate increased from 4.434×10^{13} n/s during Phase 1 to 6.616×10^{13} n/s during Phase 2. Application of Equation 1: relative change in mean neutron production therefore produced a 49.21% increase in mean neutron rate between the two operating phases.

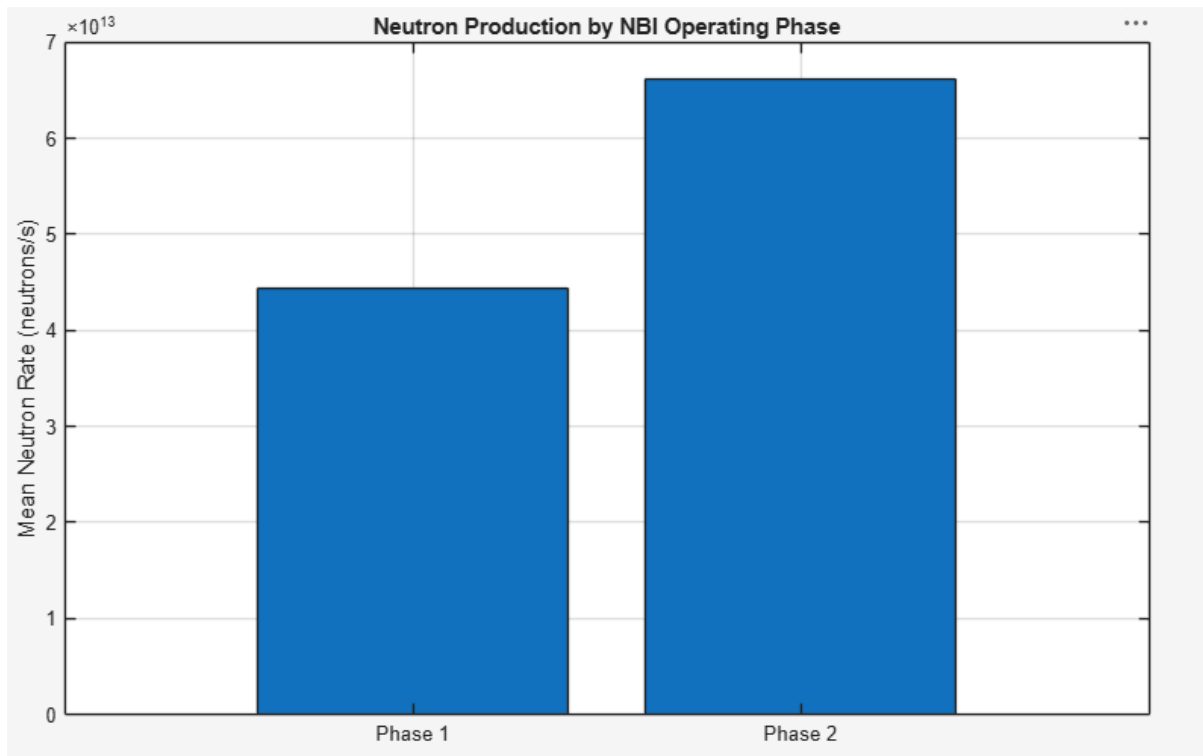


Figure 5: Mean neutron rate during the two identified NBI operating phases. Phase 1 corresponded to a mean NBI power of 1.797 MW and Phase 2 to a mean NBI power of 3.182 MW

Parameter	Phase 1	Phase 2	Change
Observations	1,441	1,006	—
Mean NBI power (MW)	1.797	3.182	+77.1%
Mean neutron rate (n/s)	4.434×10^{13}	6.616×10^{13}	+49.21%

Table 4: Comparison of the two identified NBI operating phases

Overall, the phase comparison demonstrated a clear increase in mean neutron production between the two NBI operating conditions. The increase was more clearly resolved by comparison of the phase averages than by the overall NBI neutron Pearson correlation alone. The temporal characteristics of this relationship were investigated using cross correlation and focused step response analysis, as presented in the next section Temporal Correlation and NBI Step Response.

Temporal Correlation and NBI Step Response

Cross correlation analysis was used to investigate the temporal relationships between the signals and neutron production. The plasma current and neutron rate signals exhibited a maximum normalised cross correlation coefficient of 0.6555 at zero lag, indicating that the strongest correlation between these signals occurred without a measurable temporal displacement at the 200 μ s resolution used for the analysis.

Cross correlation between NBI power and neutron rate produced a maximum coefficient of 0.7318 at a lag of -460 samples, equivalent to -92.0 ms on the common time base. The negative lag indicates that the strongest whole window alignment was obtained when the signals were displaced relative to one another in this direction. However, this value was not interpreted as a physical neutron response delay because the analysis incorporated the complete 0.00 – 0.50 s window and was therefore sensitive to the broader evolution of both signals.

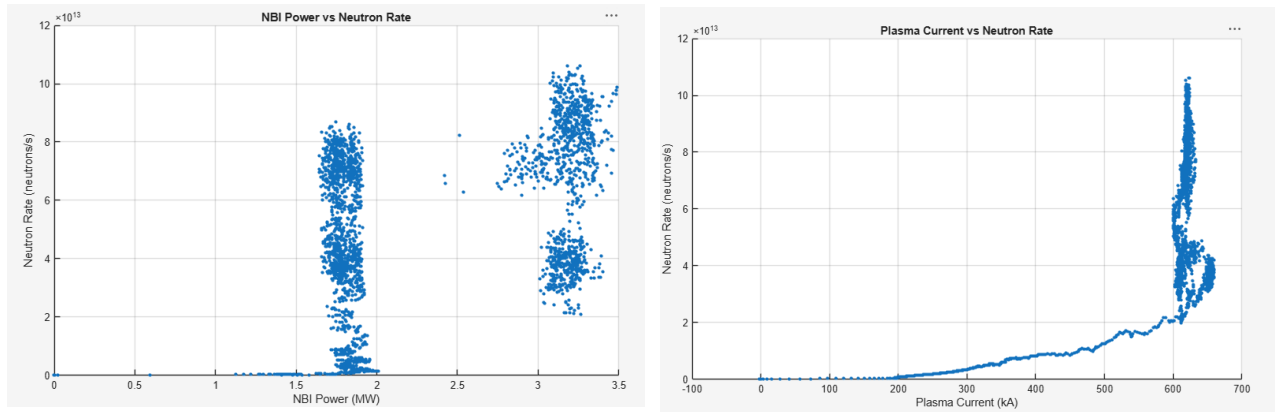


Figure 6: Normalised cross-correlation for NBI power–neutron rate (left) and plasma current–neutron rate (right), with peak correlations at -92.0 ms and 0 ms, respectively.

The difference between the zero lag plasma current relationship and the cross-correlation lag. NBI neutron result indicated that whole window cross correlation was influenced by the broader temporal structure of the signals. A focused step response analysis was therefore performed around the discrete change in NBI power to provide a more localised assessment of the neutron response.

The NBI transition was identified at 0.2980 s. Across the defined pre and post transition intervals, mean NBI power increased from 1.7945 MW to 2.9828 MW, corresponding to an increase of 66.23% . Over the same intervals, mean neutron rate increased from 6.7978×10^{13} n/s to 7.3131×10^{13} n/s, representing an increase of 7.58% .

Parameter	Pre-transition	Post-transition	Change
Mean NBI power (MW)	1.7945	2.9828	+66.23%
Mean neutron rate (n/s)	6.7978×10^{13}	7.3131×10^{13}	+7.58%
Neutron-rate SD (n/s)	3.2800×10^{12}	4.5222×10^{12}	—

Table 5: NBI step-response comparison

The pre-transition neutron rate had a mean of 6.7978×10^{13} n/s and a standard deviation of 3.2800×10^{12} n/s, producing a 3σ response threshold of 7.7819×10^{13} n/s. This threshold was first exceeded at 0.3042 s, corresponding to an apparent response delay of 6.2 ms after the NBI transition.

The neutron rate maximum reached 1.0610×10^{14} n/s at 0.3496 s, approximately 51.6 ms after the NBI transition.

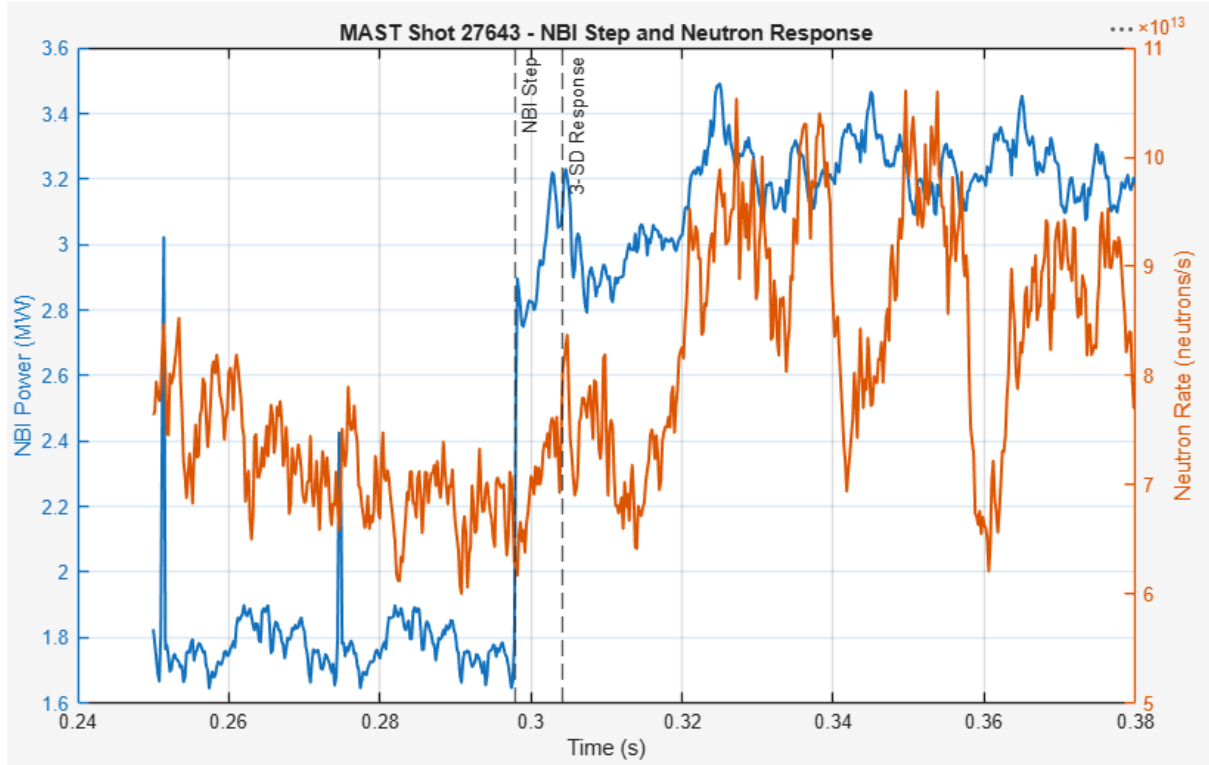


Figure 7: NBI step transition and corresponding neutron-rate response, showing the 3σ threshold and apparent 6.2 ms response delay

The analysis therefore produced an apparent response delay substantially shorter than the cross-correlation lag obtained from whole window cross correlation. These values represent different analytical measures: the cross correlation lag describes the temporal displacement producing the strongest overall statistical correspondence, whereas the 6.2 ms delay represents the first exceedance of the predefined 3σ response threshold. It is therefore treated as an apparent threshold response delay, rather than a direct measurement of an intrinsic plasma response time.

Neutron Measurement Uncertainty and Monte Carlo Analysis

Analysis of the ANU error signal showed that the absolute uncertainty associated with the neutron rate measurement varied with the magnitude of the measured signal. Across the aligned analysis window, the mean absolute uncertainty was 4.413×10^{12} n/s, with values ranging from 7.675×10^9 to 8.909×10^{12} n/s. In relative terms, the uncertainty remained comparatively stable throughout the analysis window. The mean relative uncertainty was 8.468%, with a median of 8.424% and a range of 8.397–9.018%. Mean relative uncertainty was similar between the two NBI operating phases, at 8.493% in Phase 1 and 8.421% in Phase 2.

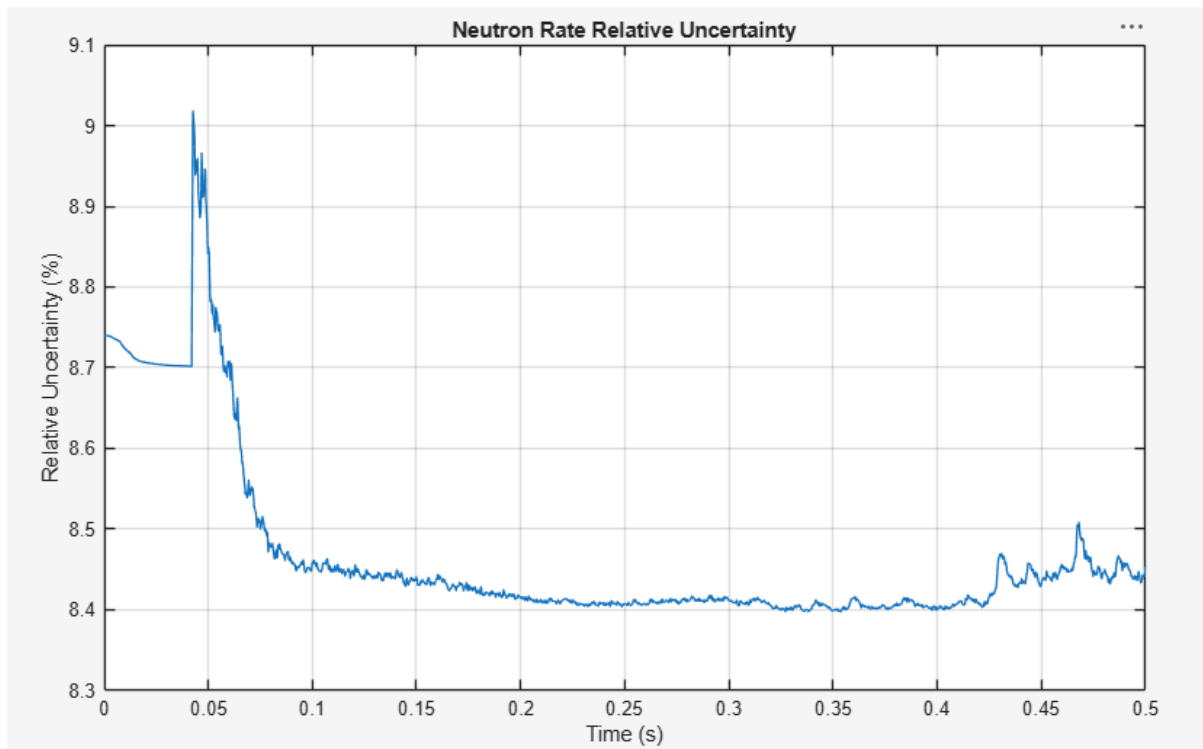


Figure 8 Relative neutron-rate measurement uncertainty across the 0.00–0.50 s analysis window

Relative uncertainty was highest during the early part of the analysis window before decreasing as neutron production increased. From approximately 0.10 s onwards, the relative uncertainty remained comparatively stable at approximately 8.4%, with only small variations during the remainder of the discharge.

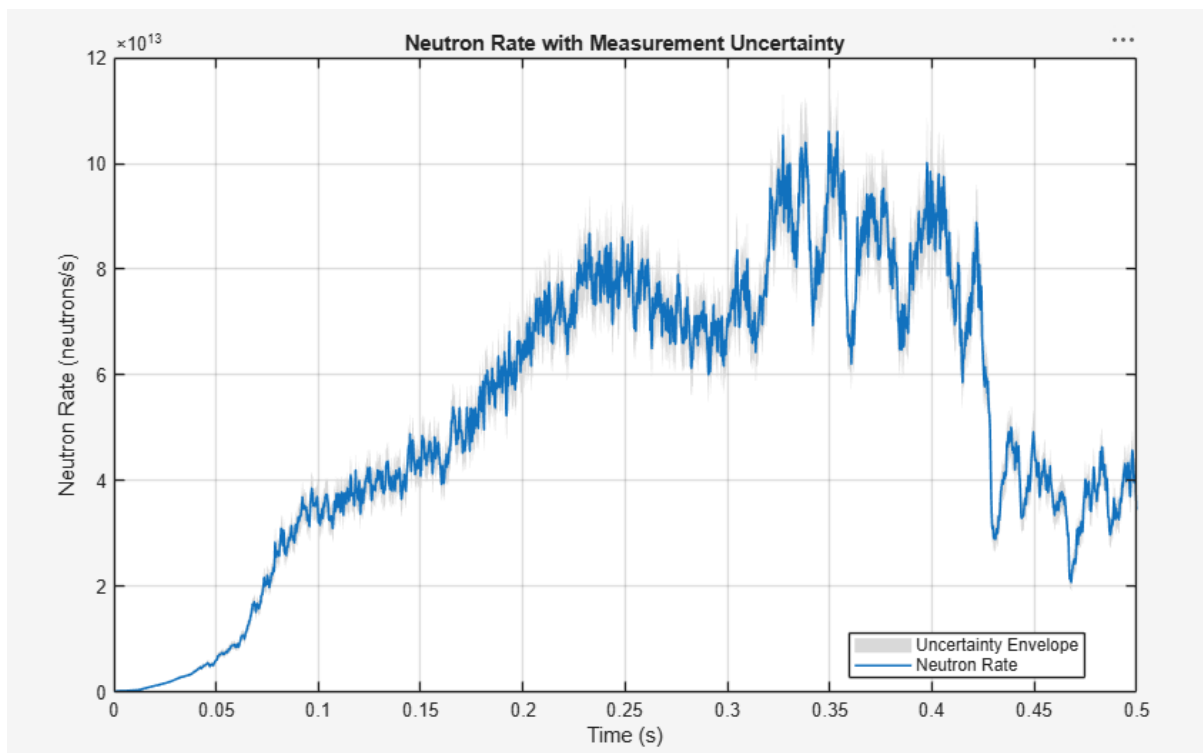


Figure 9: Neutron-rate time series with the associated measurement uncertainty envelope

Across the complete analysis window, the mean absolute uncertainty was 4.413×10^{12} n/s, while the mean and median relative uncertainties were 8.468% and 8.424%, respectively. Mean relative uncertainty was similar between the two NBI operating phases, at 8.493% for Phase 1 and 8.421% for Phase 2.

At the measured neutron rate maximum of 1.0610×10^{14} n/s at 0.3496 s, the associated absolute uncertainty was 8.9091×10^{12} n/s, equivalent to a relative uncertainty of 8.397%.

Parameter	Result
Mean absolute uncertainty	4.4132×10^{12} n/s
Median absolute uncertainty	4.4659×10^{12} n/s
Mean relative uncertainty	8.468%
Median relative uncertainty	8.424%
Relative uncertainty range	8.397–9.018%
Phase 1 mean relative uncertainty	8.493%
Phase 2 mean relative uncertainty	8.421%
Peak neutron rate	1.0610×10^{14} n/s
Peak absolute uncertainty	8.9091×10^{12} n/s
Peak relative uncertainty	8.397%

Table 6: Absolute and relative uncertainty values were derived from the ANU error signal associated with the neutron-rate diagnostic.

To evaluate the effect of the assumed measurement uncertainty on the measured peak neutron rate, an initial Monte Carlo simulation was performed using 100,000 iterations. The measured peak of 1.0610×10^{14} n/s and its associated error of 8.9091×10^{12} n/s were used as the inputs to the simulation, with the ANU error treated as a 1σ uncertainty as described in Uncertainty and Monte Carlo Analysis

The simulated distribution had a mean neutron rate of 1.0608×10^{14} n/s and a median of 1.0605×10^{14} n/s, closely corresponding to the measured value. The simulation standard deviation was 8.9251×10^{12} n/s, equivalent to 8.413% of the simulated mean. The central 95% simulation interval extended from 8.8683×10^{13} to 1.2365×10^{14} n/s.

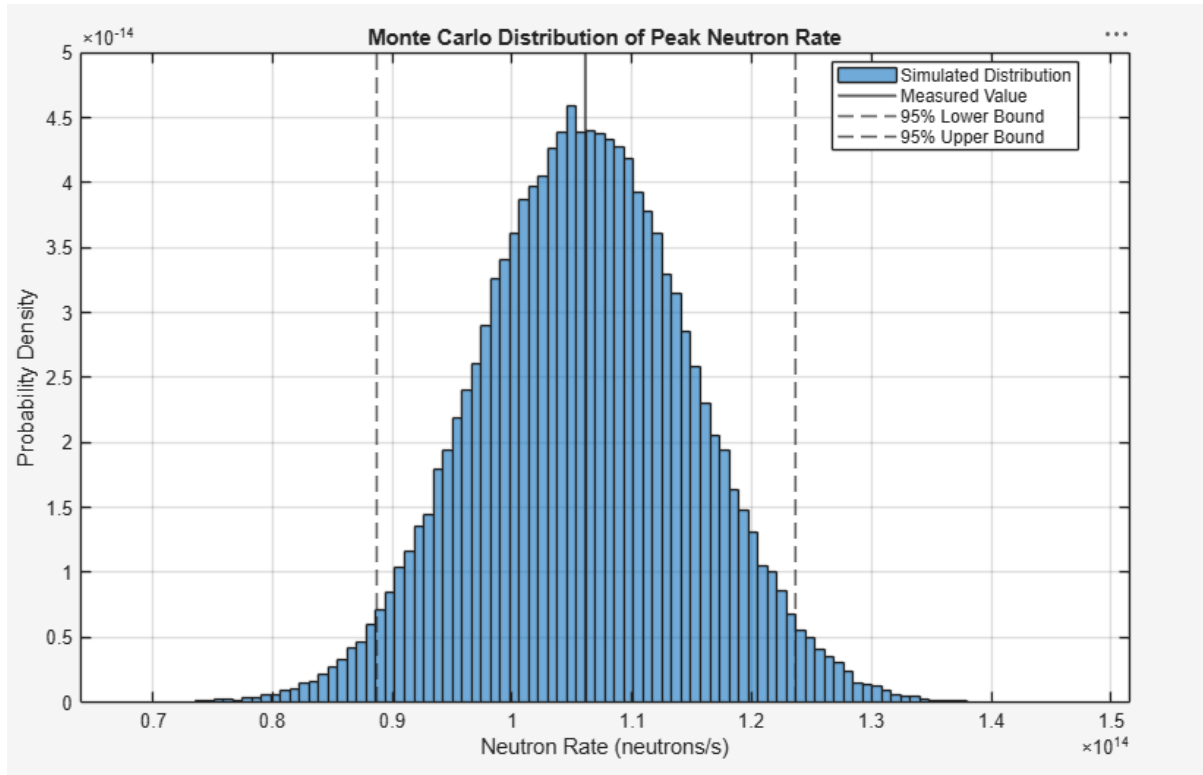


Figure 10: Monte Carlo distribution of the measured peak neutron rate with the 95% simulation interval

The simulated distribution remained centred close to the measured peak neutron rate, indicating that the Monte Carlo procedure reproduced the central value and assumed uncertainty used as its inputs. Based on the simulated distribution, the probability of the neutron rate exceeding 1.00×10^{14} n/s was 75.16%, decreasing to 32.95% above 1.10×10^{14} n/s and 6.02% above 1.20×10^{14} n/s.

The Monte Carlo analysis was extended across the complete aligned neutron rate time series using 10,000 simulated iterations. Measurement uncertainty was propagated independently at each time point using the corresponding ANU error value, producing simulated neutron rate histories across the full 0.00–0.50 s analysis window. Point by point statistics were then calculated from the simulated data to quantify the resulting uncertainty in the neutron rate time series and derived parameters.

Figure 3.11 presents the measured neutron-rate signal together with the 95% Monte Carlo simulation envelope and point-by-point simulated mean. The simulated mean closely followed the measured signal throughout the analysis window, while the width of the simulation envelope varied with the magnitude of the associated measurement uncertainty.

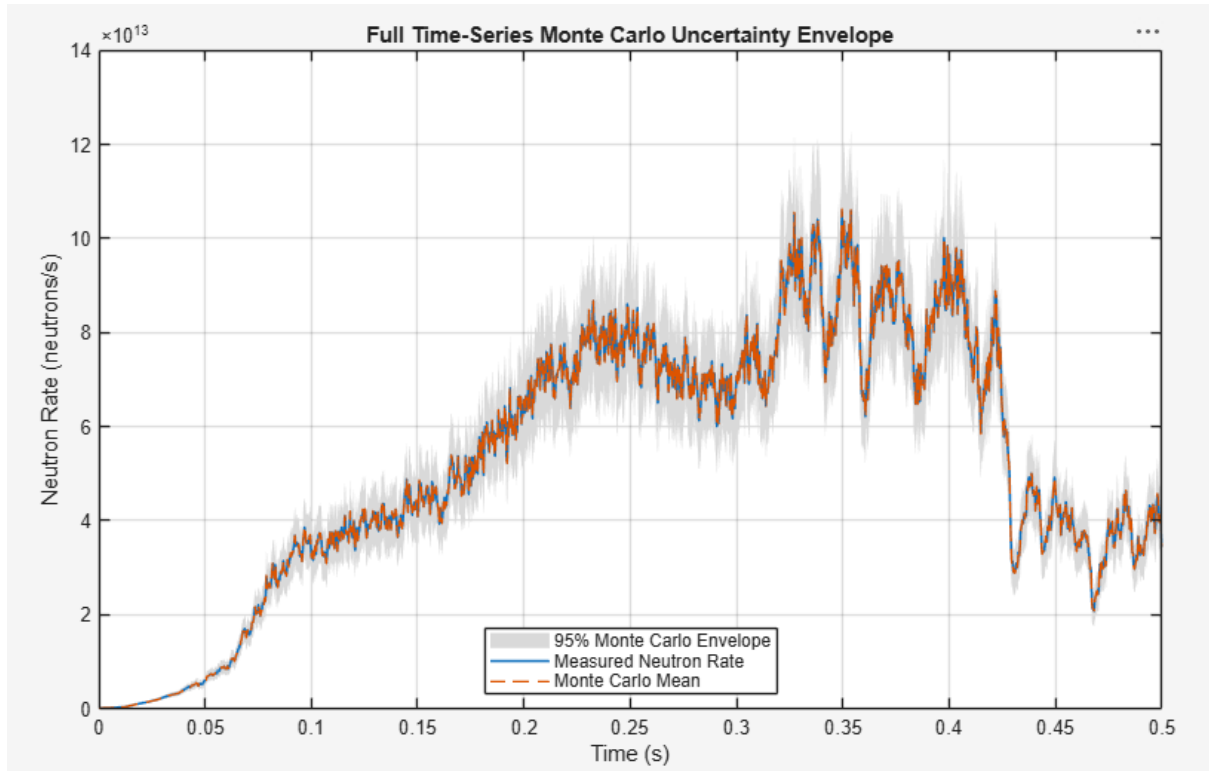


Figure 11: Measured neutron-rate time series with the point-by-point Monte Carlo mean and 95% simulation envelope.

Propagation of measurement uncertainty through the NBI phase analysis produced a mean Phase 1 neutron rate of 4.4339×10^{13} n/s, with a 95% simulation interval of 4.4109×10^{13} to 4.4567×10^{13} n/s. For Phase 2, the mean simulated neutron rate was 6.6160×10^{13} n/s, with a 95% interval of 6.5791×10^{13} to 6.6524×10^{13} n/s.

The resulting percentage increase in mean neutron rate between the two phases had a simulated mean of 49.216% and median of 49.220%. The 95% simulation interval extended from 48.094% to 50.314%.

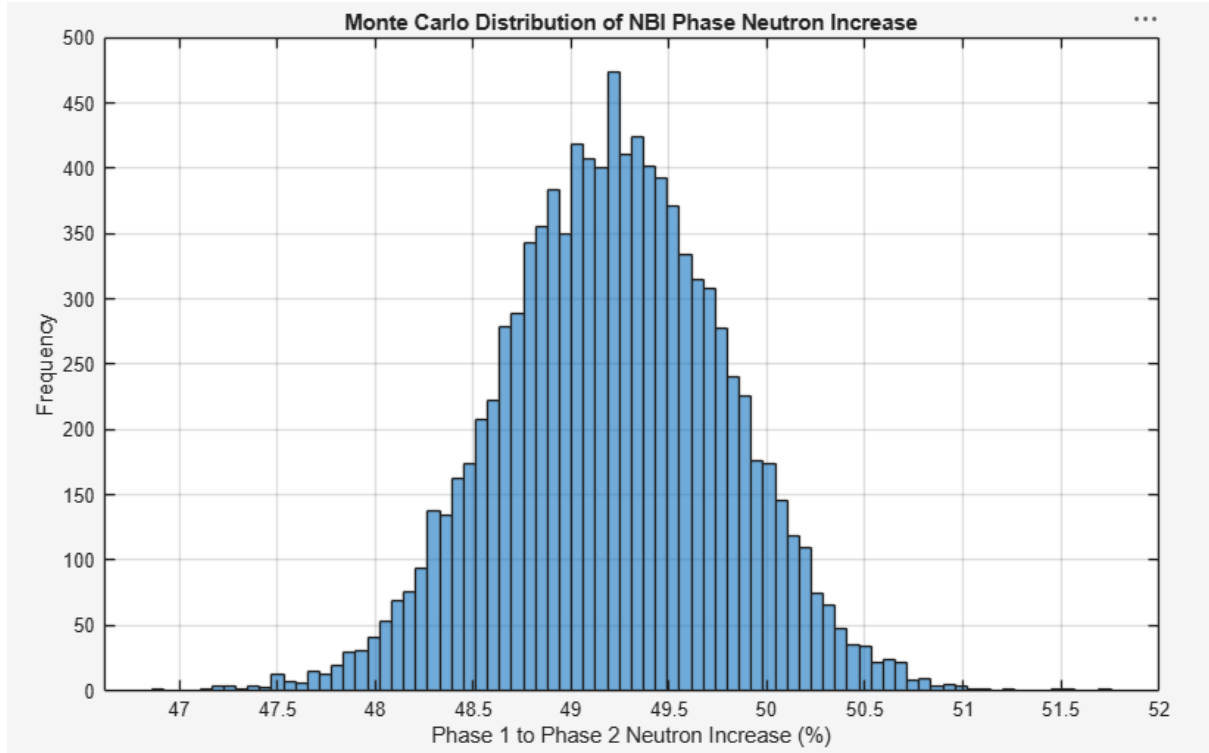


Figure 12: Monte Carlo distribution of the percentage increase in mean neutron rate from NBI Phase 1 to Phase 2.

The simulated distribution was centred close to the measured phase increase of 49.206%, with all values within the reported 95% simulation interval remaining substantially above zero. The phase comparison therefore remained consistent under propagation of the assumed neutron measurement uncertainty. The full time series simulations were also used to examine uncertainty in the magnitude and timing of the maximum neutron rate. For each simulated time series, the maximum neutron rate and its corresponding time were identified.

The mean of the simulated maximum neutron rates was 1.2089×10^{14} n/s, with a median of 1.2041×10^{14} n/s and a 95% simulation interval of 1.1446×10^{14} to 1.3014×10^{14} n/s. These values were higher than the measured maximum of 1.0610×10^{14} n/s because the analysis selected the maximum value independently from every noise-perturbed simulated time series. The resulting maximum selection effect is considered further in the Discussion

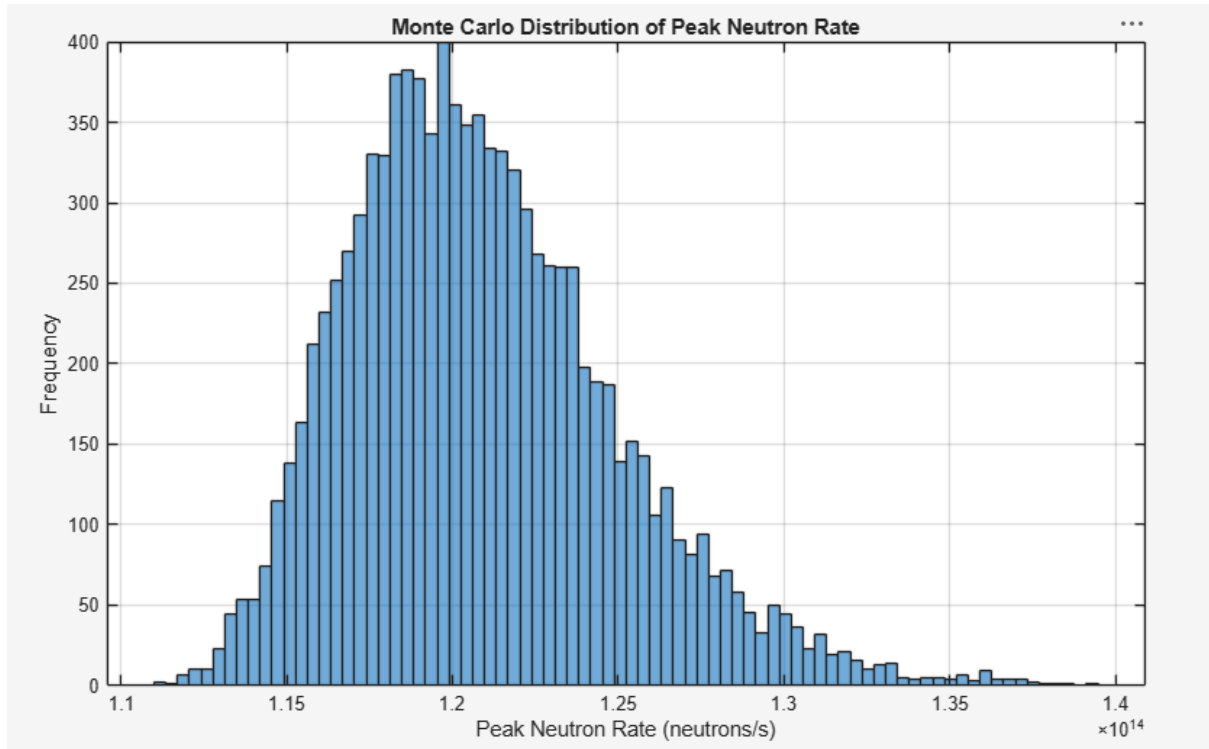


Figure 13: Distribution of maximum neutron rates obtained from the full time-series Monte Carlo simulations

The timing of the simulated maximum also exhibited substantial variation. The mean simulated peak time was 0.3466 s, while the median was 0.3494 s, compared with the measured peak time of 0.3496 s. The 95% simulation interval extended from 0.3254 to 0.4002 s, indicating that the precise location of the maximum was sensitive to measurement perturbation.

Parameter	Mean	95% simulation interval
Maximum neutron rate (n/s)	1.2089×10^{14}	$1.1446\text{--}1.3014 \times 10^{14}$
Peak time (s)	0.3466	0.3254–0.4002
Phase 1 mean neutron rate (n/s)	4.4339×10^{13}	$4.4109\text{--}4.4567 \times 10^{13}$
Phase 2 mean neutron rate (n/s)	6.6160×10^{13}	$6.5791\text{--}6.6524 \times 10^{13}$
Phase 1 to Phase 2 increase (%)	49.216	48.094–50.314

Figure 14: Summary of full time-series Monte Carlo results.

Overall, propagation of the assumed measurement uncertainty produced comparatively narrow distributions for the phase averaged neutron rates and the calculated increase between NBI operating phases. In contrast, the magnitude and timing of the maximum neutron rate were more sensitive to point-by-point measurement perturbation. The implications of these findings, together with the assumptions and limitations of the uncertainty treatment, are considered in the following Discussion

Discussion

Interpretation of Diagnostic Relationships

The analysis identified positive relationships between plasma current, NBI power and neutron production during MAST Shot 27643, although the strength of these relationships differed. The strongest pairwise association was observed between plasma current and neutron rate ($r = 0.6555$), while NBI power exhibited a more moderate relationship with neutron rate ($r = 0.4196$). These results indicate that neutron production evolved alongside both plasma current and injected beam power during the analysed discharge, but neither parameter alone described the full variation observed in neutron rate.

The positive relationship between NBI power and neutron rate is consistent with the role of neutral beam injection in generating energetic fast ions within MAST plasmas. Fast ion behaviour is important to neutron production in beam heated MAST discharges, and neutron measurements have consequently been used extensively as a diagnostic of the confined fast-ion population (Jones, 2015) This provides a physical basis for the observed tendency towards higher neutron rates under conditions of increased NBI power.

The moderate NBI neutron correlation ($r = 0.4196$), however, indicates that the relationship was not simply linear across the complete analysis window. This is consistent with previous MAST research demonstrating that neutron emission can be strongly affected by changes in fast-ion confinement and redistribution. For example, resonant magnetic perturbations have been associated with substantial reductions in neutron emission through degradation of fast-ion confinement, while energetic-particle-driven magnetohydrodynamic modes have also been shown to redistribute or cause losses of NBI generated fast ions (McClements, 2015) (Jones, 2015) Consequently, instantaneous NBI power alone would not necessarily be expected to account for all variation in neutron production.

The stronger plasma current neutron correlation ($r = 0.6555$) should likewise be interpreted cautiously. Plasma current forms part of the evolving discharge conditions and is related to plasma equilibrium and fast-ion confinement rather than acting as an isolated determinant of neutron production. MAST experiments have demonstrated that changes in plasma current and operating conditions can alter the confinement and behaviour of beam-generated fast ions; for example, RMP experiments showed substantially different neutron behaviour between lower- and higher-current plasma conditions (McClements *et al.*, 2015). The observed correlation therefore indicates related temporal evolution of plasma current and neutron production during Shot 27643, rather than demonstrating a direct causal relationship (McClements, 2015)

Taken together, the results demonstrate that neutron production during Shot 27643 was associated with both NBI operation and the evolving plasma conditions. This interpretation is consistent with previous MAST studies in which neutron measurements were sensitive to the behaviour and confinement of NBI generated fast ions (Jones, 2015) (McClements, 2015) The present analysis does not, however, establish NBI power or plasma current as independent causal predictors of neutron production. Resolving their individual contributions would require consideration of additional plasma diagnostics and, ideally, comparison across multiple discharges.

Temporal Behaviour and Response

The temporal analyses produced different results depending on whether the complete analysis window or the local NBI transition was considered. Plasma current and neutron rate exhibited maximum cross-correlation at zero lag ($r = 0.6555$), whereas NBI power and neutron rate produced a maximum cross correlation of 0.7318 at the cross-correlation lag. The latter should not be interpreted as a physical neutron response time, as whole window cross correlation identifies the temporal displacement producing the greatest statistical similarity and can be influenced by broader signal structure and changes in operating conditions.

This distinction is relevant to MAST, where neutron emission is influenced by the behaviour of NBI generated fast ions. Previous studies have demonstrated that energetic particle instabilities and changes in fast-ion confinement can redistribute fast ions and produce corresponding changes in neutron emission (Jones, 2015) (McClements, 2015) The neutron rate signal therefore cannot be expected to behave simply as a time shifted representation of NBI power.

The focused step response analysis provided a more localised measure of the behaviour surrounding the NBI transition. Following the identified NBI step at 0.2980 s, the neutron rate first exceeded the predefined 3σ threshold at 0.3042 s, giving an apparent threshold response delay of 6.2 ms. The subsequent neutron rate maximum occurred 51.6 ms after the transition. A measurable change in neutron production following altered NBI conditions is physically consistent with the importance of injected energetic deuterium ions to neutron production in beam heated MAST plasmas (Sperduti, 2020)

The cross-correlation lag and apparent threshold-response delay therefore represent different analytical quantities and should not be directly compared as estimates of the same physical response time. The former describes the broader temporal relationship between the complete signals, whereas the latter characterises the local neutron response following a specific NBI transition. The 6.2 ms value is consequently interpreted as an apparent threshold response delay, rather than an intrinsic plasma time constant.

Robustness of Derived Metrics

The Monte Carlo analysis indicated that the phase averaged neutron response was more robust than the peak based measurements. The measured increase in mean neutron rate between Phase 1 and Phase 2 was 49.21%, compared with a simulated mean of 49.216% and a narrow 95% simulation interval of 48.094–50.314%. This demonstrates that the observed increase remained stable under the modelled measurement uncertainty.

In contrast, the magnitude and timing of the neutron rate maximum were more sensitive to measurement perturbation. The mean simulated maximum (1.2089×10^{14} n/s) exceeded the measured maximum (1.0610×10^{14} n/s) due to the statistical effect of selecting the highest value from each simulated time series. Peak timing also showed greater variability, with a 95% interval of 0.3254–0.4002 s.

Overall, the phase to phase increase represents the more robust derived result, while the precise magnitude and timing of the neutron rate maximum should be interpreted with greater caution.

Limitations and Assumptions

Several limitations should be considered when interpreting the results. The analysis was based on a single MAST discharge, and the observed relationships should therefore not be assumed to represent behaviour across other plasma conditions or discharges. In addition, interpolation to a common 200 μ s time base and selection of the 0.00–0.50 s analysis window introduced analytical choices that may influence the resulting statistical relationships.

The temporal results are also dependent on the selected analytical methods. Pearson correlation and cross correlation quantify statistical relationships but do not establish causality, while the 6.2 ms apparent response delay depends on the chosen 3σ threshold and should not be interpreted as an intrinsic plasma response time.

A further limitation concerns the ANU error data. The available metadata did not define whether ANU_ERRORS represented a standard deviation or another uncertainty measure. For Monte Carlo analysis, these values were therefore assumed to represent 1σ normally distributed uncertainties, with errors treated as independent between time points. Correlated and systematic uncertainties were not modelled.

Consequently, the results should be interpreted as an analysis of Shot 27643 under the stated assumptions, with broader conclusions requiring analysis of additional MAST discharges and further characterisation of the diagnostic uncertainty model.

Conclusion

Analysis of MAST Shot 27643 identified a clear relationship between neutral beam injection (NBI) conditions and neutron production. Mean NBI power increased from 1.797 MW to 3.182 MW between the two identified operating phases, while mean neutron rate increased by 49.21%, from 4.434×10^{13} to 6.616×10^{13} n/s. The neutron response was therefore substantial but sub linear relative to the increase in NBI power, consistent with neutron production being influenced by the wider plasma state and fast ion behaviour rather than instantaneous beam power alone.

Temporal analysis further showed that whole window and localised methods described different aspects of the discharge. Although NBI power and neutron rate produced a maximum cross correlation at –92 ms, focused analysis of the NBI transition identified an apparent threshold response delay of 6.2 ms. The latter provides evidence of a localised change in neutron behaviour following the NBI transition but is not interpreted as an intrinsic plasma response time.

The neutron rate uncertainty derived from the supplied ANU error data was approximately 8.4%, comparable in magnitude to the approximately 8% uncertainty previously reported for calibrated MAST neutron yield measurements (Stammers, 2006). Monte Carlo propagation was used to determine how this measurement uncertainty affected the principal derived results. Despite the uncertainty associated with individual neutron rate measurements, the measured 49.21% increase in mean neutron rate between the two NBI phases remained highly stable, with a simulated mean increase of 49.216% and a 95% simulation interval of 48.094–50.314%. This demonstrates that, under the adopted uncertainty model, the observed phase to phase increase was substantially greater than could be explained by measurement uncertainty alone. In contrast, the magnitude and timing of individual neutron rate peaks

were more sensitive to measurement perturbation, demonstrating that phase averaged behaviour provided a more robust basis for interpretation than individual peak measurements.

Overall, the analysis demonstrates that the neutron response in Shot 27643 cannot be characterised by NBI power alone. The observed sub-linear response and dependence on wider discharge evolution provide a strong basis for further multidagnostic investigation, particularly of fast ion confinement, plasma conditions and MHD activity, to determine the mechanisms underlying the observed neutron behaviour.

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Appendix A — Dataset and Variable Details

Diagnostic	Signal	NetCDF file	NetCDF variable/group	Units
AMC	Plasma current	amc27643.nc	/amc/plasma_current/data	kA
ANB	NBI power	anb27643.nc	/anb/tot_sum_power/data	MW
ANU	Neutron rate	anu27643.nc	/anu/neutrons/data	neutrons/s
ANU	Neutron error	anu27643.nc	/anu/errors/data	error, neutr

Table 7: NetCDF diagnostic data sources used for Shot 27643

Signal	Original signal name	Description	Original observations	Native median timestep
Plasma current	AMC...	Plasma current	30,000	200 μ s
NBI power	ANB...	Neutral beam injection power	50,001	50 μ s
Neutron rate	ANU_NEUTRONS	Neutron rate	46,704	10 μ s
Neutron uncertainty	ANU_ERRORS	Supplied neutron error signal	46,704	10 μ s

Table 8: Diagnostic signal metadata

Appendix B — Raw Data Inspection and Quality Assessment

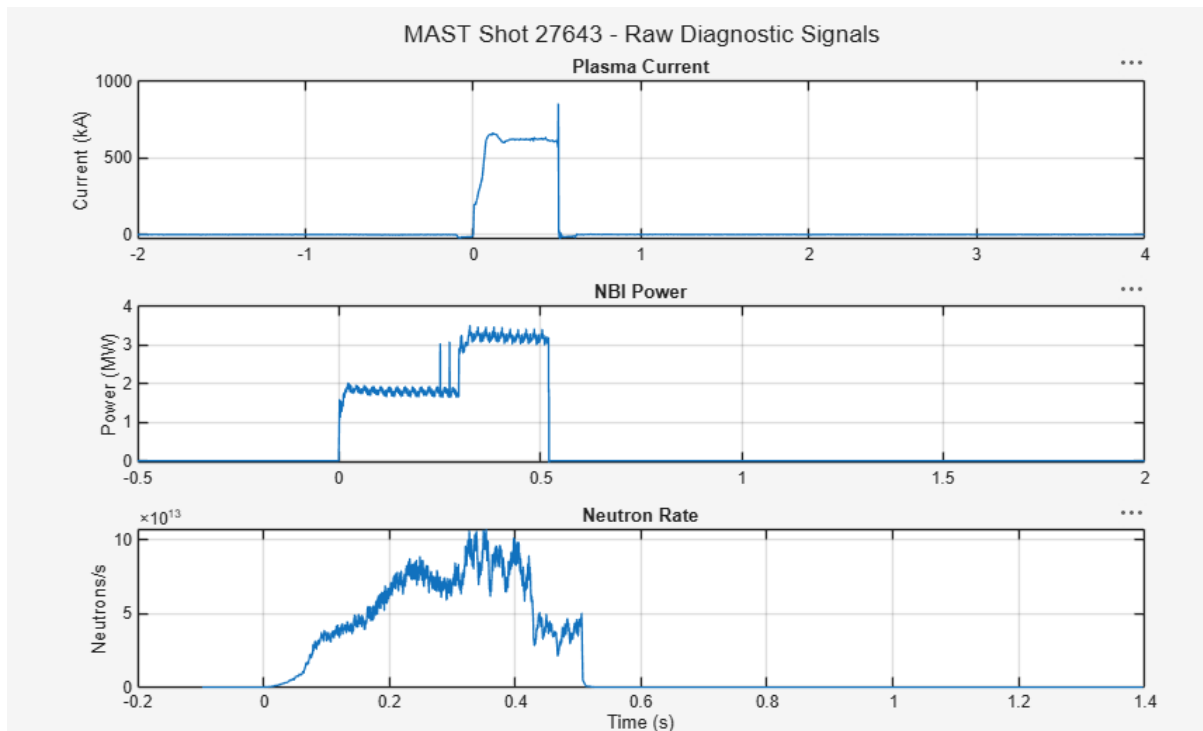


Figure 15: Raw plasma-current, NBI-power and neutron-rate signals for MAST Shot 27643 prior to restriction to the analysis window

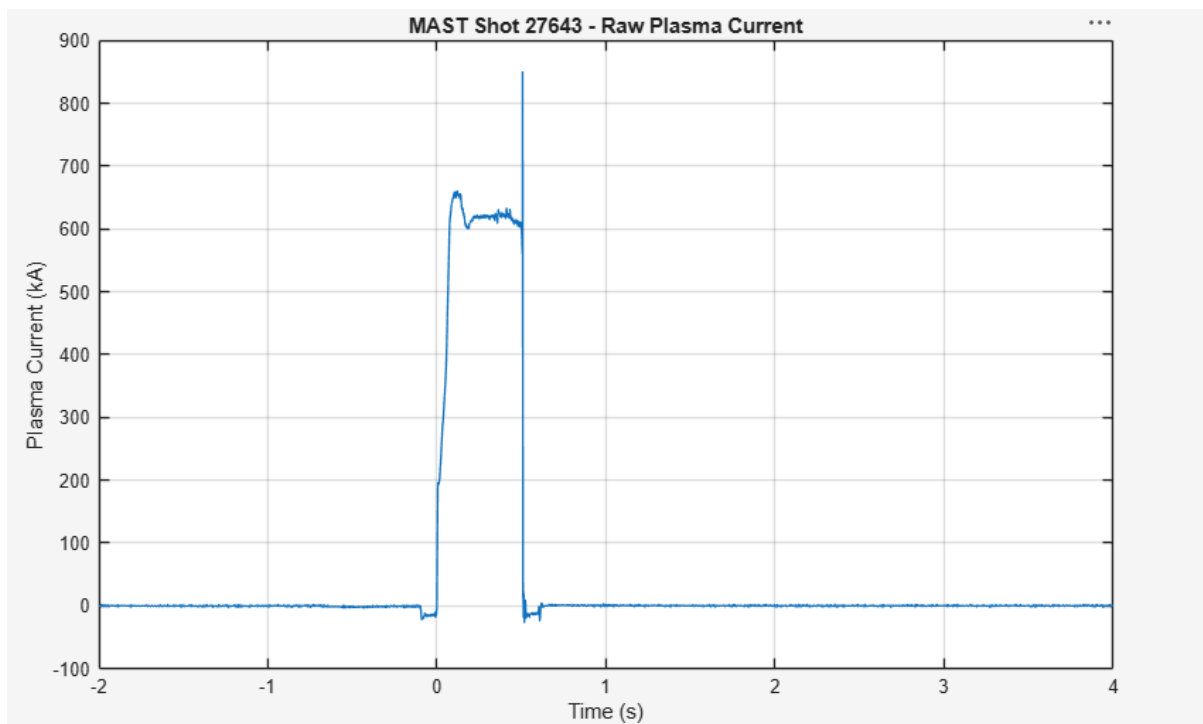


Figure 16 Raw plasma-current signal for Shot 27643.:

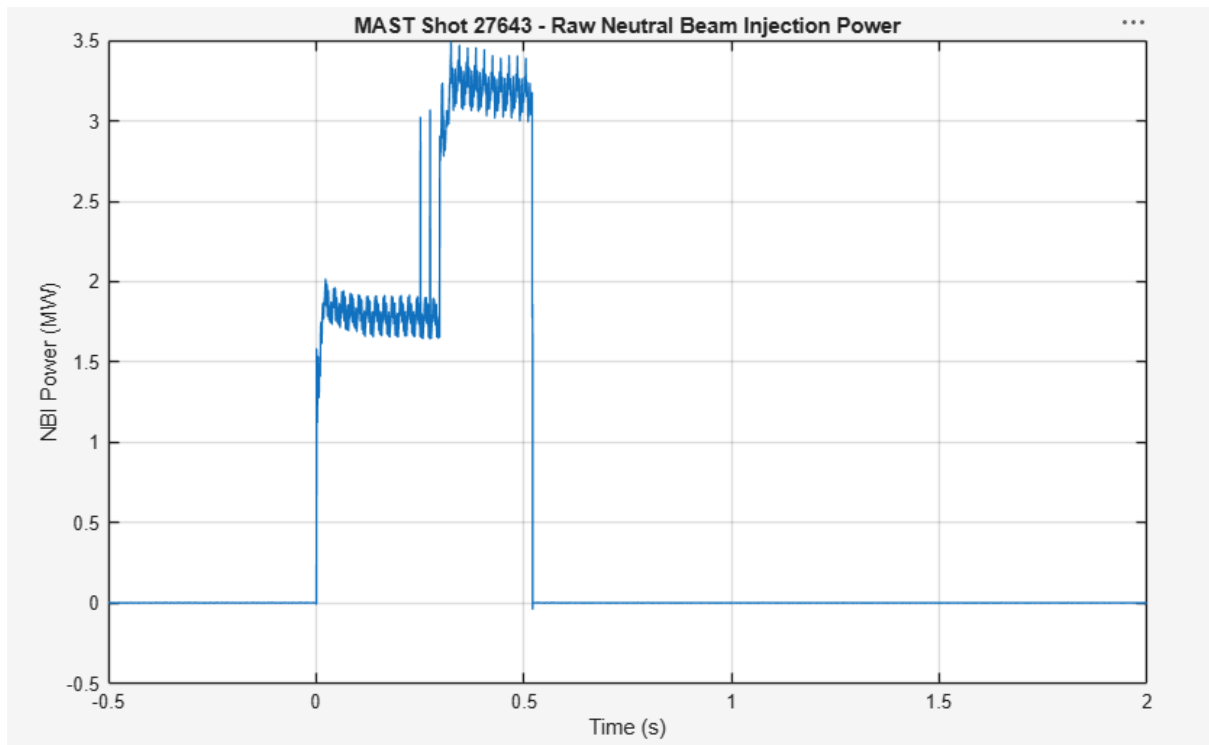


Figure 17: Raw NBI-power signal for Shot 27643.

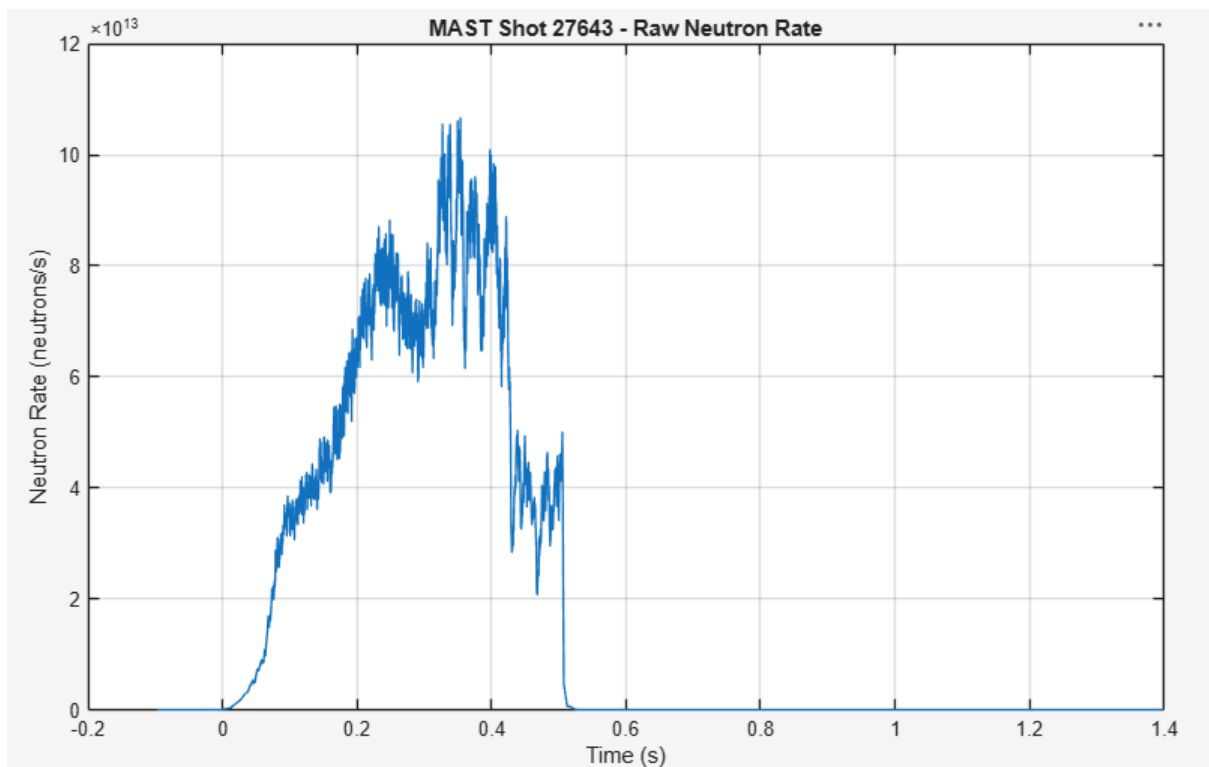


Figure 18: Raw neutron-rate signal for Shot 27643.

Appendix C — Supplementary Statistical and Temporal Analysis

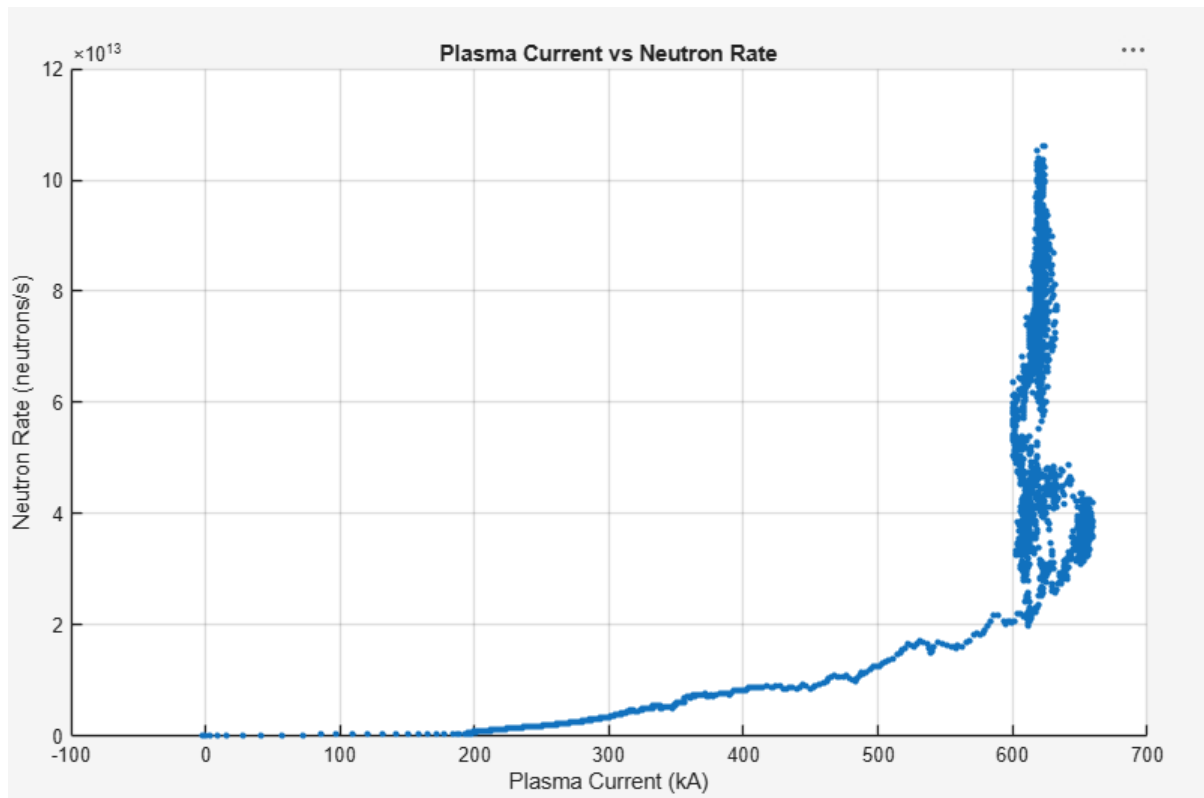


Figure 19: Scatter plot of aligned plasma current against neutron rate.

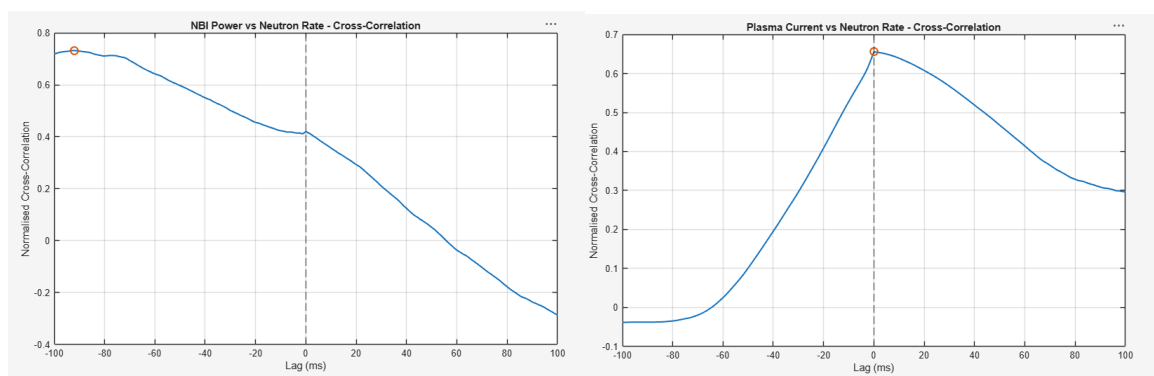


Figure 20: Normalised cross-correlation of (a) NBI power and neutron rate and (b) plasma current and neutron rate.